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In [1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
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1. For this exercise we will use the Tips dataset from the Seaborn Python library, which contains 244 observations with 7 columns - total_bill, tip, sex, smoker, day, time, size. The questions we will explore are: What is the distribution of total bills, and does it vary by day of the week? And how does the tip amount relate to the total bill, and is this impacted by smoker status?

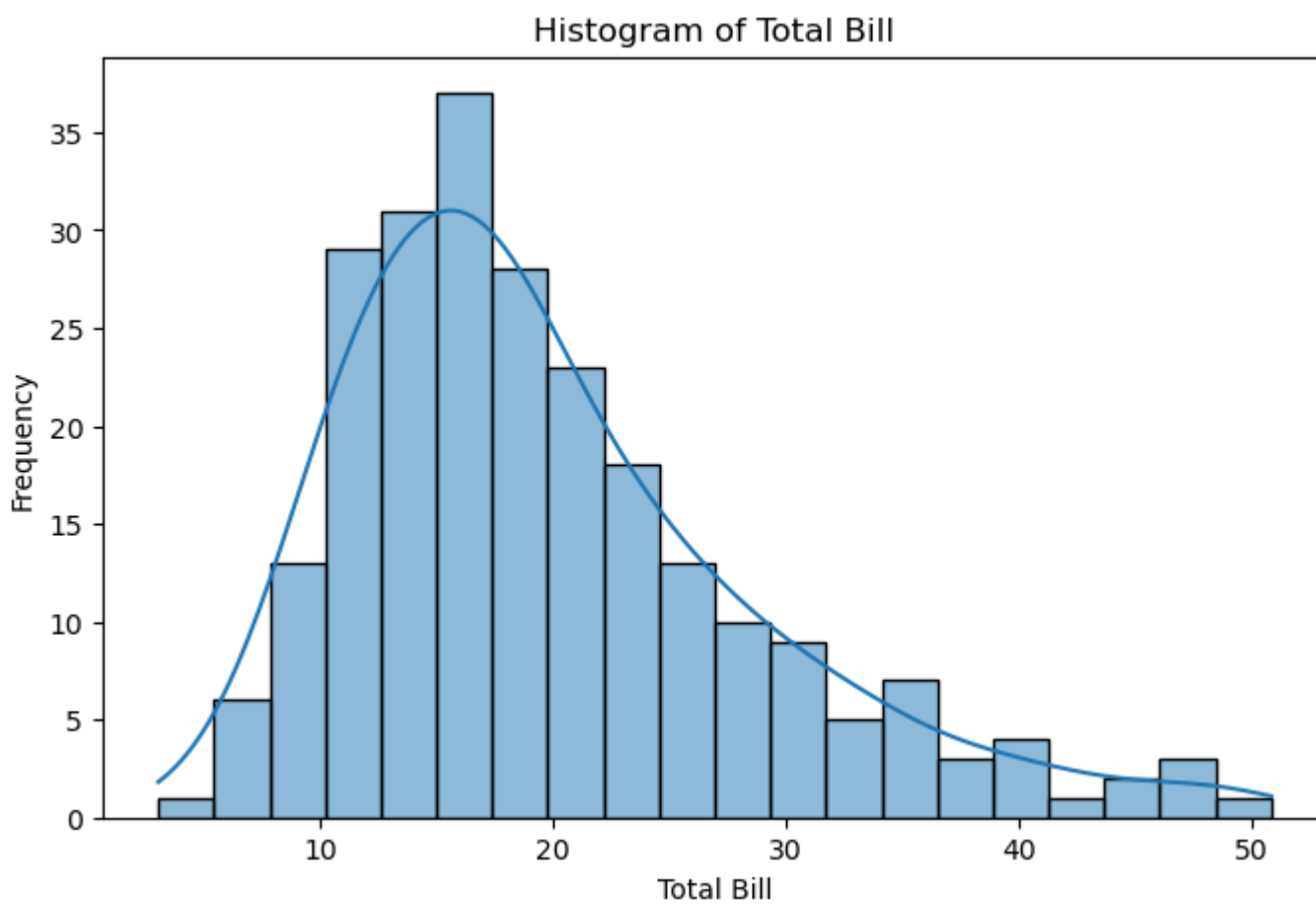
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In [2]: # Load tips dataset from Seaborn library
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tips = sns.load_dataset('tips')
print(tips.head())
print(tips.describe())
```

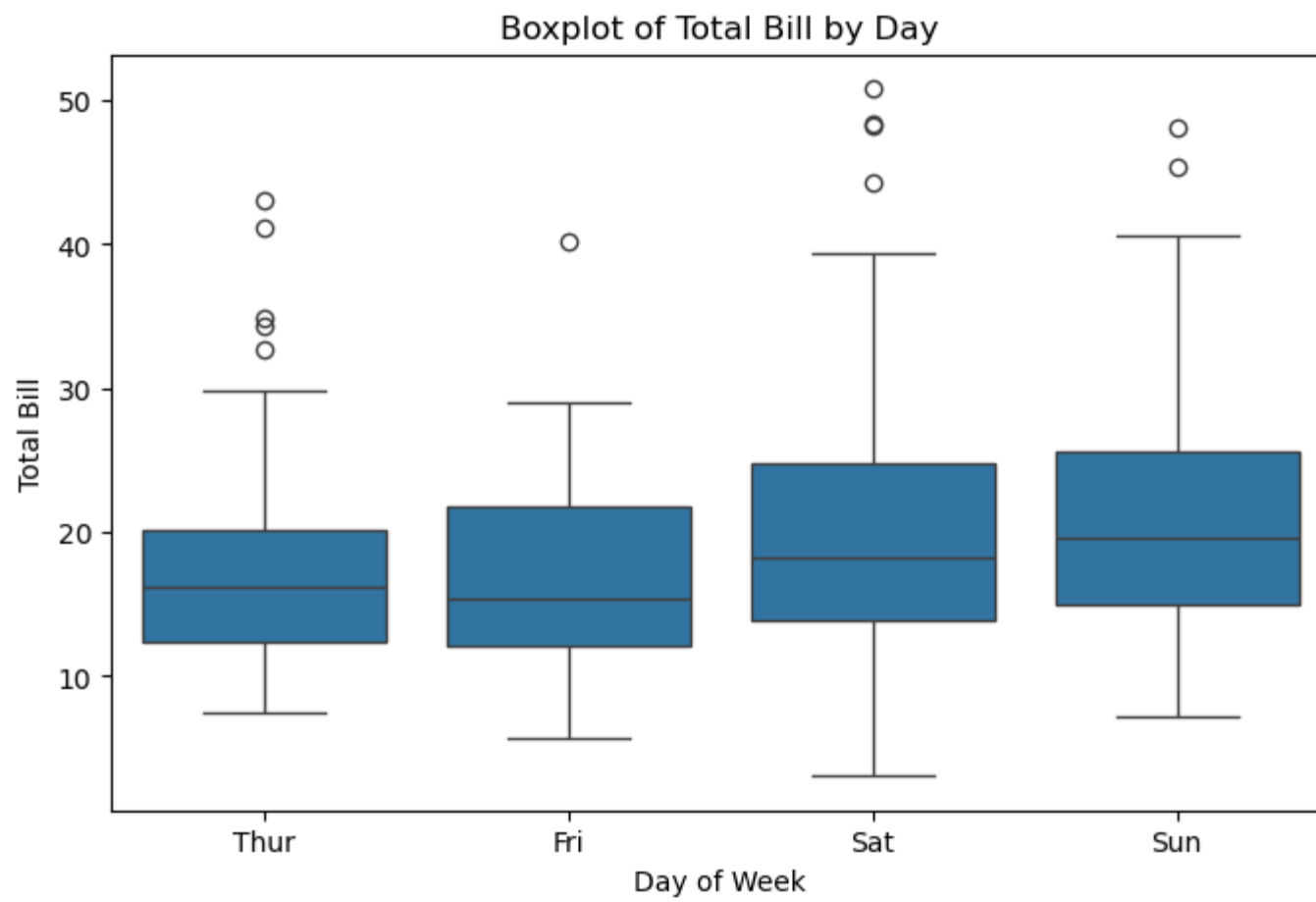
	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

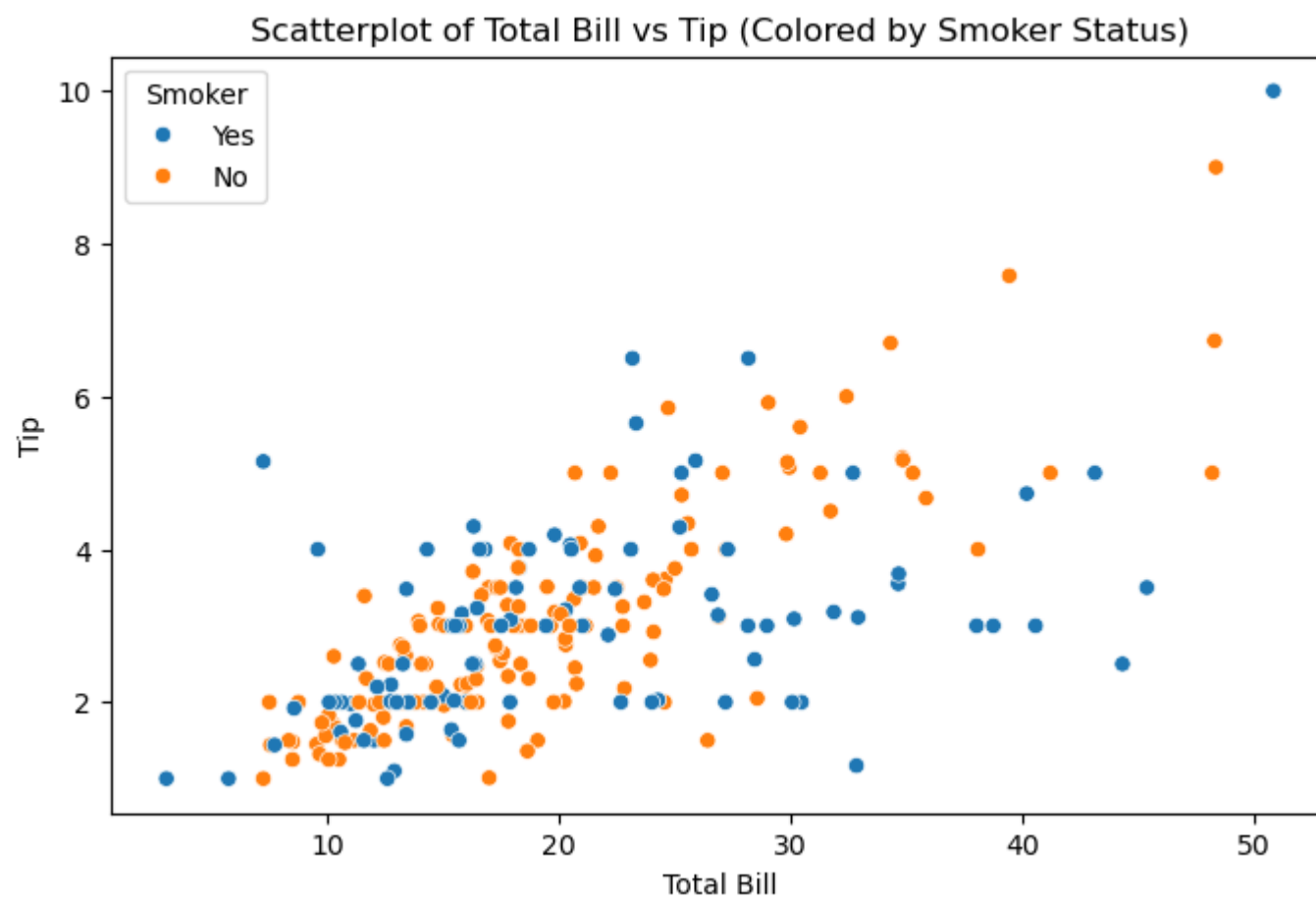
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In [3]: # 2. Histograms / Bar Graph
plt.figure(figsize=(8, 5))
sns.histplot(tips['total_bill'], bins=20, kde=True)
plt.title('Histogram of Total Bill')
plt.xlabel('Total Bill')
plt.ylabel('Frequency')
plt.show()
```



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In [4]: # 3. Boxplot
plt.figure(figsize=(8, 5))
sns.boxplot(x='day', y='total_bill', data=tips)
plt.title('Boxplot of Total Bill by Day')
plt.xlabel('Day of Week')
plt.ylabel('Total Bill')
plt.show()
```



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In [5]: # 4. Bivariate Plot (Scatterplot)
plt.figure(figsize=(8, 5))
sns.scatterplot(x='total_bill', y='tip', hue='smoker', data=tips)
plt.title('Scatterplot of Total Bill vs Tip (Colored by Smoker Status)')
plt.xlabel('Total Bill')
plt.ylabel('Tip')
plt.legend(title='Smoker')
plt.show()
```



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In [6]: # 5. Additional Visualizations - Bar plot: Count of Meals by Day
plt.figure(figsize=(8, 5))
sns.countplot(x='day', data=tips)
plt.title('Count of Meals Served by Day')
plt.xlabel('Day of Week')
plt.ylabel('Count')
plt.show()
```

